

QUANTITATIVE CHEMISTRY – TITRATIONS

KEY VOCABULARY

Titration: a method to find the exact volume for neutralisation.

Burette: measures the volume of solution added accurately.

End point: when the indicator just changes colour.

Concordant: repeat titres within 0.10 cm³ of each other.

COMMON MISCONCEPTIONS

~~Use universal indicator for titration.~~

✓ Use a single-colour-change indicator (e.g. phenolphthalein).

~~Average every titre value.~~

✓ Only average concordant results.

~~Read the burette from the top.~~

✓ Read from the bottom of the meniscus at eye level.

CONCENTRATION (MOL/DM³)

$$\text{moles} = \text{concentration} \times \frac{\text{volume (cm}^3\text{)}}{1000}$$

CONVERSION REMINDER

$$1000 \text{ cm}^3 = 1 \text{ dm}^3$$

1 APPARATUS

Name the piece of apparatus used to **add acid accurately**. [1 mark]

2 INDICATOR

Explain why a single indicator (not universal) is used in titrations. [2 marks]

3 MOLES OF ACID

25.0 cm³ of 0.10 mol/dm³ HCl is used. **Calculate** the moles of HCl. [2 marks]

Answer: _____ mol

4 FIND CONCENTRATION

0.0025 mol of NaOH is in 20.0 cm³. **Calculate** its concentration. [2 marks]

Answer: _____ mol/dm³

5 FILL THE GAPS

Complete the sentence using the word bank. [2 marks]

Read the burette at the bottom of the _____ and only average the _____ titres.

concordant

meniscus

average

top

6 ACCURACY

Suggest why a pipette is used to measure the alkali into the flask. [1 mark]